

**Milestone Two**

**Software Design and Engineering Artifact**

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The artifact for software design and engineering is a weight-tracking app originally created in Android Studio using Java. The app was built to let the user create an account, log in, add weight, update and delete records, save goal weights, and optionally enable sms settings. In this application, we see the use of SQLite to store user records, weight history, goal weight information, and local settings.

I selected this artifact for my ePortfolio because it is a working application that has a clear purpose. It is not just a small code sample because it includes screens, user interaction, persistent storage, and logic that connects the user goal to the application behavior. This makes it a good example of my ability to build software that is organized around a user need. The app also fits my ePortfolio because it shows mobile development experience, Java programming, database use, validation, and secure coding improvements.

The artifact was improved in several ways. One of the main improvements was adding safer password handling. In the original version, the password handling was weaker because the app could store or compare passwords in a way that was not safe for a real application. In the enhanced version, I added a PasswordUtils class that uses PBKDF2 password hashing with salts. This means the raw password does not have to be stored directly in the database. This was an important improvement because user account information should be treated as private information, even in a class project.

Another improvement was adding centralized validation through the WeightValidator class. This made the code easier to maintain because the rules for usernames, passwords, phone numbers, and weight values are not scattered across the whole project. The enhanced app checks for reasonable weight ranges and stronger password requirements. This supports better data quality and also makes the app less likely to accept bad or unrealistic input.

I also improved the goal weight logic and the user safety of the app. The goal weight feature now uses the logged-in user ID when checking the saved goal weight, which keeps the goal connected to the correct user. I also added a delete confirmation before a weight entry is removed. This matters because weight history is user-created data, and deleting it by mistake would be frustrating. The SMS portion was also improved so it is optional and permission based. If the user does not grant SMS permission, the app should still continue working for the main purpose of tracking weight.

In Module One we identified some course outcomes that our enhancements will align with. Now I will show how the enhancements I have completed to this point will show which course outcomes I have completed. For my initial enhancements, I have met course outcomes 2,4,5 and made progress towards course outcomes 3 and 1. I made the least progress on course outcome 1 because this project was not the strongest artifact for that outcome. The app does help an individual user make decisions about weight progress, but it is not as strong for collaborative decision-making as my database dashboard artifact will be. Course outcome 2 has been met through the initial code review, which enabled me to share a narrative describing the original project and the enhancements I would implement.

Course outcome 3 had progress been made because the enhancements required design choices. Course Outcome 4 is demonstrated by my use of Android Studio, Java, SQLite, Android activities, SharedPreferences, and Android permission handling to improve a working computing solution. Course Outcome 5 is demonstrated by improvements in password security, validation, and permission behavior.

An update on my original plan I did not complete full Room and ViewModel conversion for this milestone. Instead, I focused on practical changes that improved the app while keeping

the project stable. I added repository-style organization, stronger validation, password hashing, safer delete behavior, and corrected goal logic. I still think a full Room and MVVM structure would be a good future improvement, but the current enhancement was a solid step forward for the milestone.

This process of reviewing projects that I have worked on in the past has taught me that the original app had the main features working but when I reviewed it more closely I found areas where the design could be improved, like in the security of the app. This is a good stepping stone because now I am looking at it as a soon to be professional and will need to think about how to optimize designs and create more secure and safe programs. Part of the learning process was looking at the code from a viewpoint from a nother developer, how would someone else be able to debug this and that's when it was clear to me that separating responsibilities would make it cleaner and easier to understand. By adding helper classes for password handling and validation.

The enhancements made to the weight tracking application make it a strong ePortfolio artifact. Though this is my initial enhancement, I will continue to improve this program for my final submission.